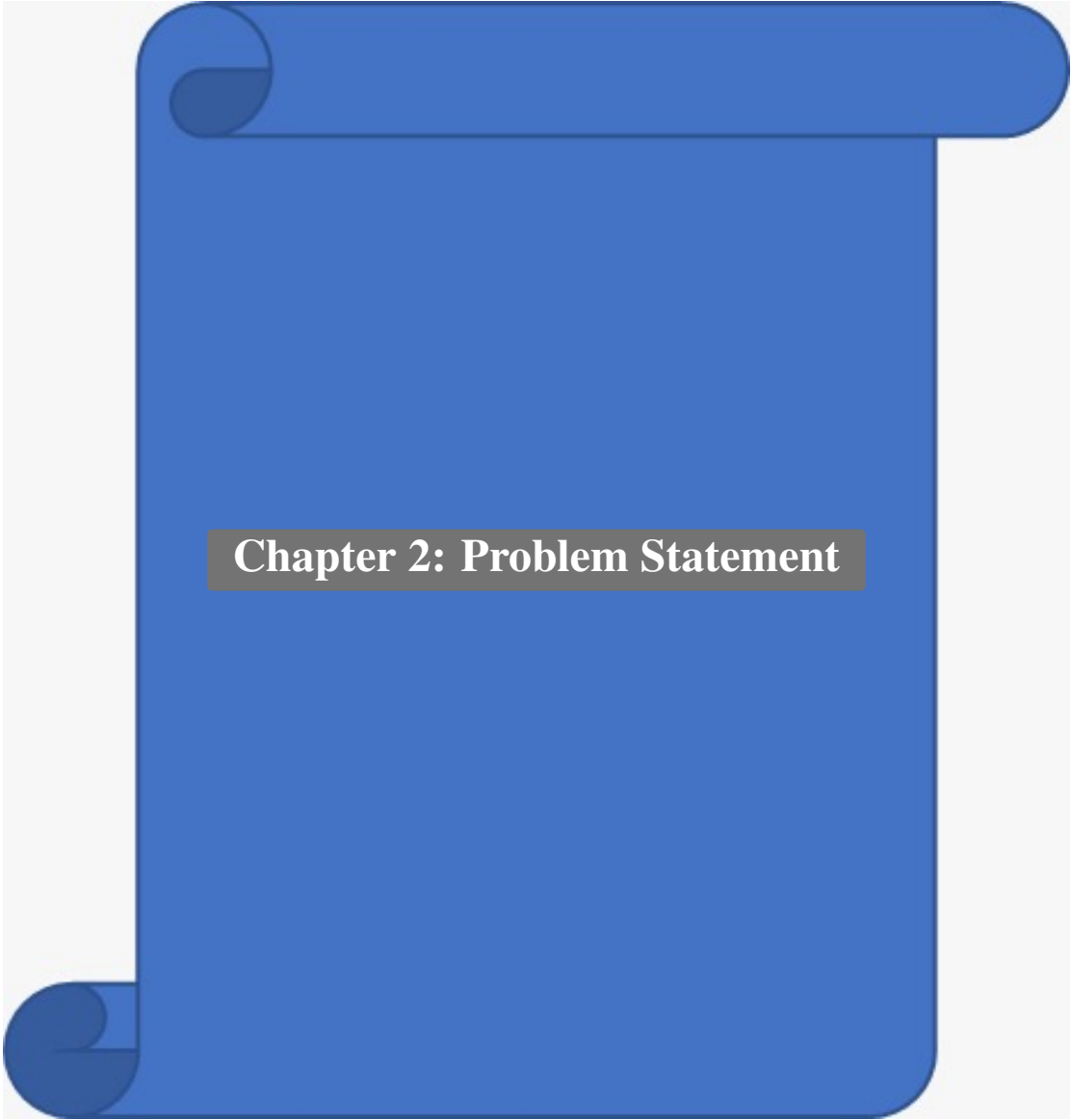




Chapter 2: Problem Statement

India generates approximately 500 million tonnes of agricultural waste annually. A substantial portion of this waste, particularly in states like Punjab, Haryana, and Uttar Pradesh, is burned in open fields due to the lack of economically viable and logistically feasible alternatives. This practice results in:

- **Air Pollution:** Stubble burning contributes to severe air quality deterioration, with PM 2.5 levels reaching hazardous levels in the Indo-Gangetic plains.
- **Health Hazards:** Respiratory diseases, cardiovascular issues, and allergic reactions increase significantly during the burning season.
- **Soil Degradation:** Burning destroys beneficial soil microorganisms, reduces organic carbon content, and diminishes soil fertility over time.
- **Greenhouse Gas Emissions:** Agricultural burning releases CO₂, CO, CH₄, and N₂O, contributing to climate change.

2.1 State-wise Crop Residue Generation

The following table presents state-wise crop residue generation and estimated residue burned, based on data from the Indian Agricultural Research Institute (IARI) and the Ministry of Agriculture:

Table 2.1: State-wise Crop Residue Generation and Burning Estimates

State	Residue Generated (MT)	Residue Burned (MT)	% Burned
Punjab	51.0	19.8	38.8
Haryana	27.6	9.1	33.0
Uttar Pradesh	59.0	21.3	36.1
Maharashtra	46.5	7.4	15.9
Madhya Pradesh	33.2	4.3	12.9
Karnataka	28.4	3.8	13.4
Andhra Pradesh	25.1	2.5	10.0
Tamil Nadu	19.8	3.9	19.7
Others	209.4	20.0	9.5
Total	500.0	92.1	18.4

2.2 Limitations of Existing Solutions

Several initiatives have been undertaken to address agricultural waste management, but they face significant limitations:

1. **Government Subsidy Programs:** While the government provides subsidies for farm machinery (e.g., Happy Seeder, Super Straw Management System), the adoption rate remains low due to high upfront costs, limited availability, and lack of awareness. Additionally, these programs focus on in-situ management rather than creating a market for waste.
2. **Manual Collection Systems:** Traditional manual collection is labor-intensive, time-consuming, and lacks transparency. There is no mechanism for tracking waste from the point of collection to the recycling facility, leading to inefficiencies and potential mismanagement.
3. **Biomass Power Plants:** Although biomass power plants can utilize agricultural waste, farmers face challenges in transporting waste to these facilities. The absence of a digital platform for coordinating logistics results in poor utilization of available capacity.
4. **Existing Agricultural Apps:** Current agricultural apps (such as Kisan Suvidha, mKisan, etc.) focus primarily on crop management, weather forecasting, and market prices. None of them provide an integrated solution for waste tracking, collection coordination, and environmental impact measurement.

The core problem is the absence of an integrated digital platform that can:

1. Enable farmers to easily report and offer their waste for collection instead of burning it.
2. Provide a transparent marketplace connecting waste generators with waste processors.
3. Track the complete lifecycle of agricultural waste from field to recycling facility.
4. Quantify the environmental impact of waste diversion in terms of carbon credits.
5. Ensure secure and transparent financial transactions for all stakeholders.
6. Provide real-time analytics and oversight for organizational administrators.

AgriTrace addresses each of these gaps by providing a unified, role-based platform with real-time tracking, integrated payments, carbon credit quantification, and comprehensive analytics.

AgriTrace addresses this problem by creating a role-based web platform that digitizes and streamlines the entire agricultural waste management pipeline.